

ELEC 221 Lecture 03

DT impulse response and convolution sum; CT convolution integral

Thursday 17 September 2026

Announcements

- Class back in person on Tuesday
- Quiz 2 on Tuesday
- Monday tutorial: tutorial assignment 1 (intro to Python)

Last time

$$x_1(t) \rightarrow \boxed{S} \rightarrow y_1(t) \quad x_2(t) \rightarrow \boxed{S} \rightarrow y_2(t)$$

We defined **LTI** (linear, time-invariant) systems.

$$ax_1(t) \rightarrow \boxed{S} \rightarrow ay_1(t)$$

Linearity:

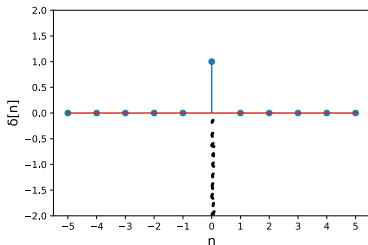
$$ax_1(t) + bx_2(t) \rightarrow \boxed{S} \rightarrow ay_1(t) + by_2(t)$$

Time invariance:

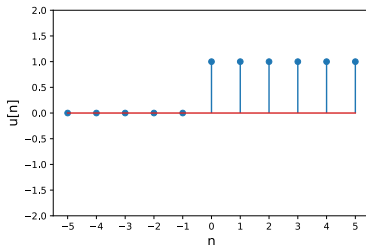
$$\begin{aligned} x(t) &\rightarrow \boxed{S} \rightarrow y(t) \\ x(t-t_0) &\rightarrow \boxed{S} \rightarrow y(t-t_0) \end{aligned}$$

Last time

We defined the DT unit impulse and unit step



$$\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$$



$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

We expressed them in terms of each other.

$$\delta[n] = u[n] - u[n-1]$$

$$\begin{aligned} u[n] &= \sum_{m=-\infty}^n \delta[m] \\ &= \sum_{k=0}^{\infty} \delta[n-k] \end{aligned}$$

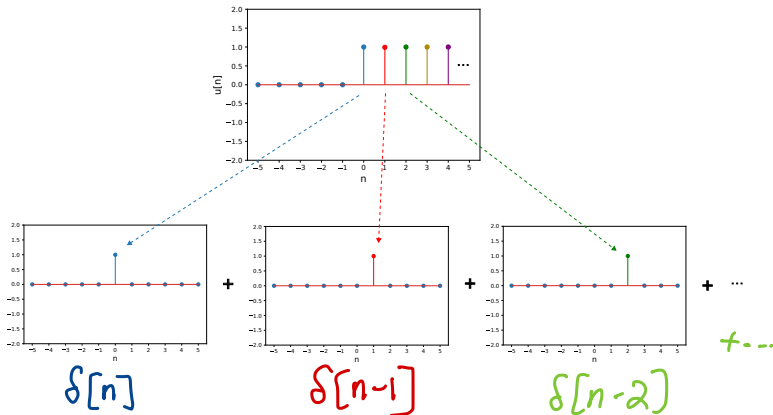
Today

Learning outcomes:

- Define the DT impulse response
- Define the convolution sum and use it to compute the output of a system
- Define the CT unit impulse and step functions
- Define the convolution integral and use it to compute the output of a system

Signals as superpositions of weighted, shifted impulses

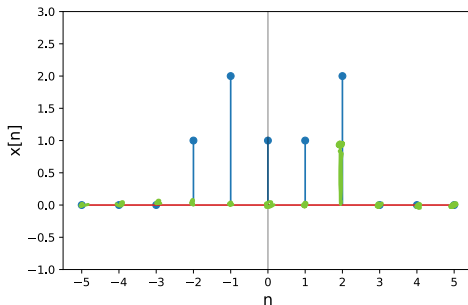
$$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$$



The unit impulse as a sampler

Multiplying by a shifted impulse “samples” the signal at that point:

$$x[n] = \delta[n+2] + 2\delta[n+1] + \delta[n] + \delta[n-1] + 2\delta[n-2]$$



$$x[n] \cdot \delta[n-k] = x[k] \cdot \delta[n-k]$$

$k=2$

The impulse response

$$x_0 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + \dots$$

How does an LTI system respond to a signal

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k]$$

$$= \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix}$$

analogous decomposition.

$$x[n] \rightarrow [S] \rightarrow y[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \underbrace{\text{system}[\delta[n-k]]}_{???$$

Suppose it sends $\delta[n-k] \rightarrow h_k[n]$

$$\delta[n-k] \rightarrow [S] \rightarrow h_k[n]$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h_k[n]$$

$h_k[n]$ is called the **impulse response**.

Real-world example: nerve conduction study

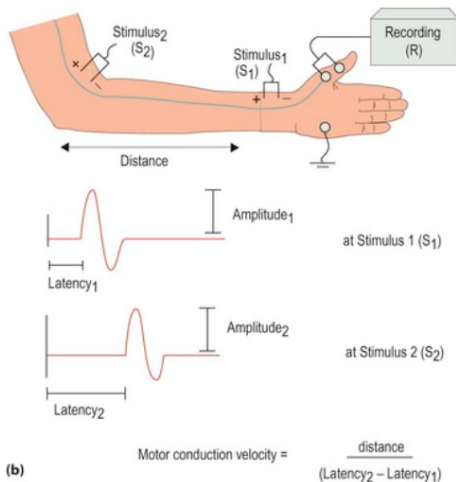


Image source: <https://neupsykey.com/nerve-conduction-studies-and-electromyography/>

The impulse response and time-invariance

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \underbrace{h_k[n]}_{\text{system output of } \delta[n-k]}$$

What if the system is also time invariant?

$$x[n] \rightarrow y[n] \quad x[n-k] \rightarrow y[n-k]$$

Then

$$\begin{aligned} \delta[n] &\rightarrow h[n] \\ \delta[n-k] &\rightarrow h[n-k] \end{aligned}$$

The convolution sum

time-invariant

If we know how a **linear** system responds to the unit impulse, we can learn how it responds to **any other signal!**

$$\delta[n] \rightarrow h[n]$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \quad (\text{input signal})$$

$$\Rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k] \quad (\text{output signal})$$

This is the **convolution sum**. We are “convolving” the sequences $x[n]$ and $h[n]$.

$$y[n] = x[n] * h[n]$$

Exercise: impulse response

$$\delta[n] \rightarrow [S] \rightarrow h[n]$$

Consider an LTI system with input/output relationship

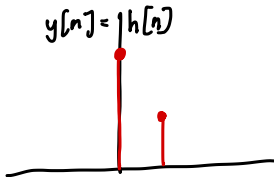
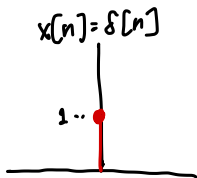
$$y[n] = 2x[n] + x[n-1]$$

What is the impulse response of the system?

$$x[n] \rightarrow [S] \rightarrow y[n] = 2x[n] + x[n-1]$$

$$\delta[n] \rightarrow [S] \rightarrow h[n] = ??$$

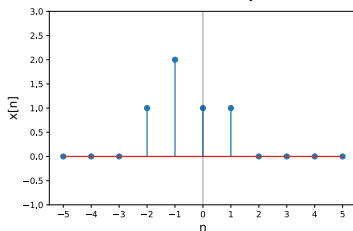
$$= 2 \cdot \delta[n] + \delta[n-1]$$



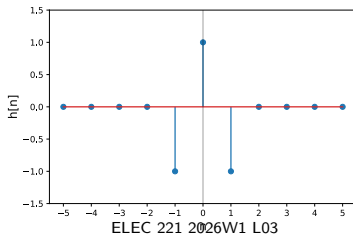
Example: convolution sum

Consider the signal

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$



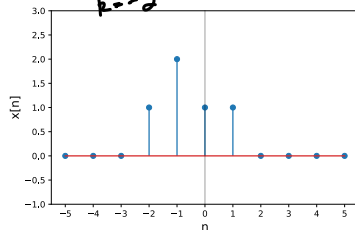
input to a system with impulse response



Example: convolution sum

To learn the system output, we must consider the contribution of each weighted impulse response:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$
$$= \sum_{k=-2}^1 x[k] h[n-k]$$



Only $x[k] \neq 0$ only for $k \in \{-2, -1, 0, 1\}$. So need to determine $x[k]h[n-k]$ for these cases, and sum them.

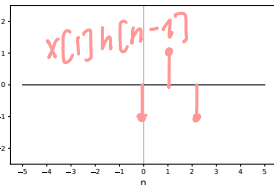
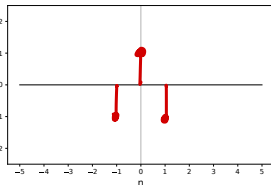
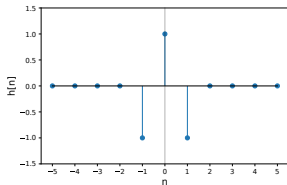
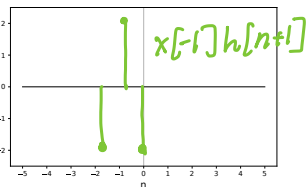
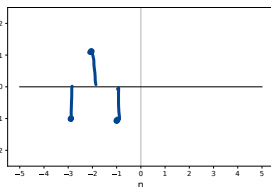
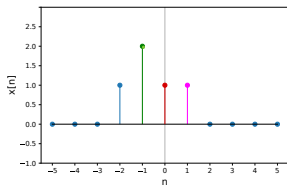
Example: convolution sum

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

$$k = -2 \Rightarrow h[n+2]$$

$$x[-1] = 2$$

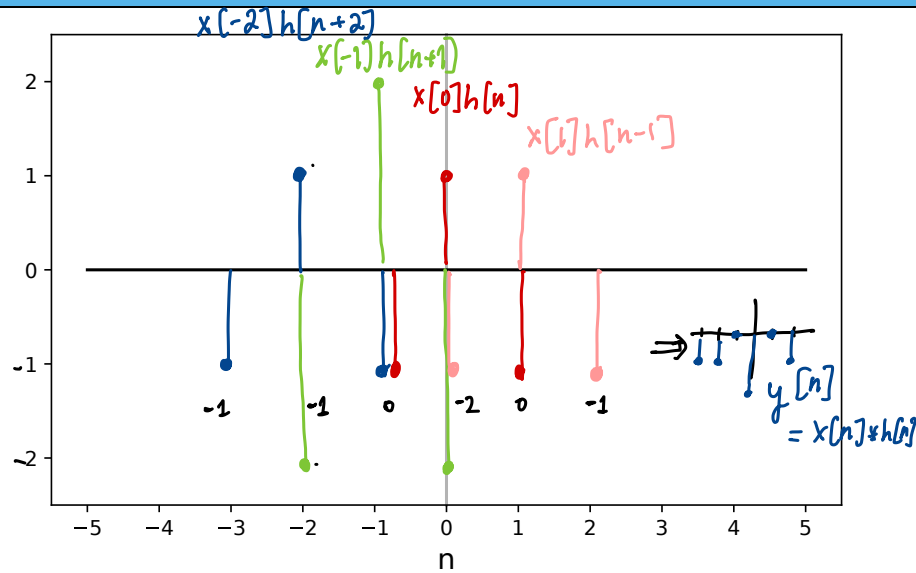
$$k = -1 \quad h[n+1]$$



$$k=0$$

$$k=1$$

Example: convolution sum



Properties of convolutions

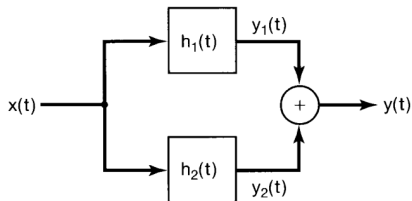
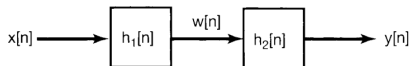


Image credits: Signals and Systems 2nd ed., Oppenheim

Convolution is:

- Associative:

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

- Commutative:

$$x[n] * h[n] = h[n] * x[n]$$

- Distributive:

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

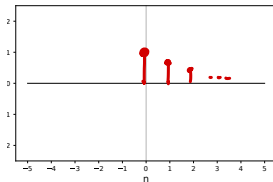
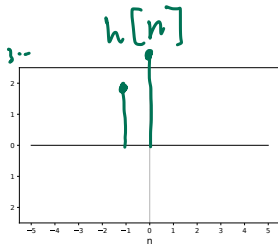
Example: convolution sum

Consider an LTI system with impulse response

$$h[n] = 3\delta[n] + 2\delta[n+1]$$

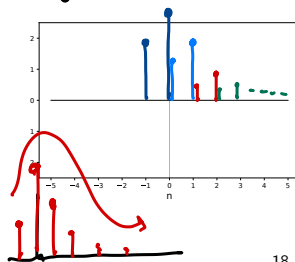
What is output of the system if

$$x[n] = \left(\frac{2}{3}\right)^n u[n]$$



Qualitatively:
 $y[n]$ $n < 0$: $h[0]$ is 2 at $n = -1$
 impulse at $n = 1$: smaller response (scaled by $\frac{2}{3}$)

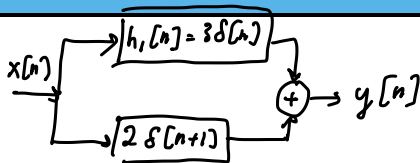
$$y[n] = x[n] * h[n]$$



$$x[0] = 1 \quad x[1] = \frac{2}{3}$$

Example/exercise: convolution sum

What is output of the system



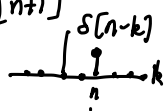
$$x[n] = \left(\frac{2}{3}\right)^n u[n], \quad h[n] = 3\delta[n] + 2\delta[n+1]$$

$$\begin{aligned} y[n] &= x[n] * h[n] = x[n] * (3\delta[n] + 2\delta[n+1]) \\ &= 3x[n] * \delta[n] + 2x[n] * \delta[n+1] \end{aligned}$$

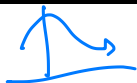
$$\begin{aligned} 3x[n] * \delta[n] &= 3 \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] = 3 \sum_{k=-\infty}^{\infty} \left(\frac{2}{3}\right)^k u[k] \cdot \delta[n-k] \\ &= 3 \sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k u[k] \delta[n-k] \\ &= 3 \cdot \left(\frac{2}{3}\right)^n u[n] \end{aligned}$$

which terms (which k) survive the sum? $\Rightarrow k=n$

still need?



Example/exercise: convolution sum

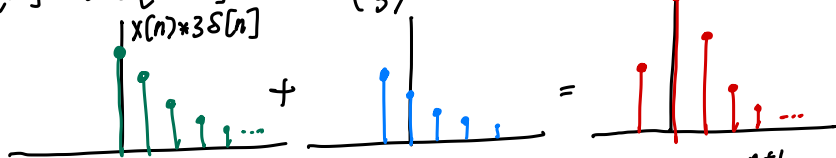


$$x[n] * 3\delta[n] = 3\left(\frac{2}{3}\right)^n u[n]$$

What is output of the system

$$x[n] = \left(\frac{2}{3}\right)^n u[n], \quad h[n] = 3\delta[n] + 2\delta[n+1]$$

$$x[n] * 2\delta[n+1] = 2 \cdot \left(\frac{2}{3}\right)^{n+1} u[n+1]$$



$$y[n] = x[n] * h[n] = 3\left(\frac{2}{3}\right)^n u[n] + 2 \cdot \left(\frac{2}{3}\right)^{n+1} u[n+1]$$

For next time

Content: *CT unit impulse + convolution integral*
unit impulse + system char.

- ~~(Complex) exponential signals~~
- ~~The continuous time Fourier series representation~~

Action items:

- ① Study for quiz 2 on Tuesday
- ② Check PrairieLearn for practice assignment problems

Recommended reading: *see L2 for more accurate list.*

- From today's class: Oppenheim 1.4, 2.1-2.3
- practice problems: 2.1-2.12, 2.14-16, 2.21, 2.22, 2.28, 2.29
- For next class: Oppenheim 1.3, 3.0-3.3